Deep Generative Models

Chapter 4: Generative Adversarial Networks

Ali Bereyhi

ali.bereyhi@utoronto.ca

Department of Electrical and Computer Engineering

University of Toronto

Summer 2026

Low-dimensional Latent

From our discussions on latent space: *we expect that the latent be of much lower dimension* $m \ll d$

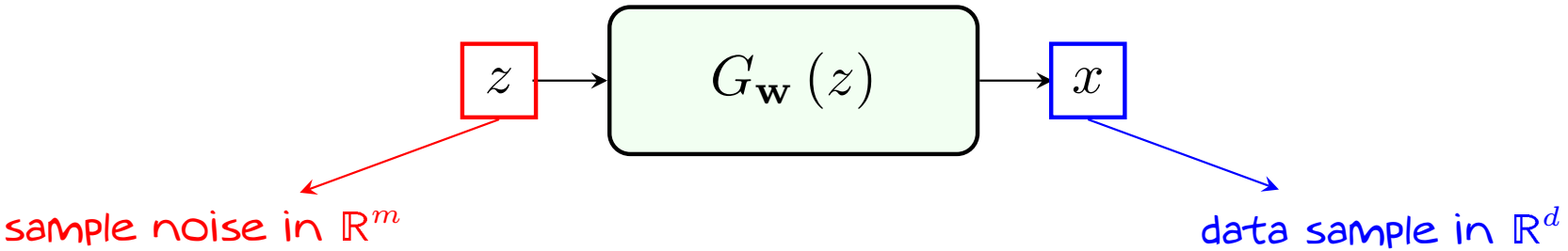
- ! *We could not work with such latent in flow-based models*
  - ↳ *The flow needs to be invertible*
- 💡 *We however intuitively expect that a much smaller latent could do the job*

- \+ *What if we try mapping a low-dimensional latent into the data space?*
- \+ *Can't we use simply MLE?!*
- – *Not explicitly! Let's take a look*

Generator: *Non-invertible Flow*

Generator Model

Generator model $G_{\mathbf{w}} : \mathbb{R}^m \mapsto \mathbb{R}^d$ *is a mapping that maps a latent sample* $z \sim Q(z)$, *typically distributed Gaussian, into a data sample*



+ *Isn't that what we called flow?!*
– Not exactly! *Flow is invertible*

Generator Distribution: *Example*

Consider a dummy example: $z \in \mathbb{R}$ *and* $x \in \mathbb{R}^3$ *with generator*

$$G(z) = \left[-z^2, \exp\{-z\}, z\right]$$

Say, want to find the CDF at $x = [x_1, x_2, x_3]$*: we need to find*

$$\begin{aligned}\mathbb{Z}(x) &= \{z \in \mathbb{R} : G(z) \leqslant x\} \\ &= \{z \in \mathbb{R} : -z^2 \leqslant x_1 \text{ and} \\ &\qquad \exp\{-z\} \leqslant x_2 \text{ and} \\ &\qquad z \leqslant x_3\}\end{aligned}$$

Generator Distribution: *Example*

1. *First constraint requires*

$$-z^2 \leqslant x_1 \rightsquigarrow z^2 \geqslant -x_1 \rightsquigarrow \begin{cases} z \geqslant \sqrt{-x_1} \\ z \leqslant -\sqrt{-x_1} \end{cases}$$

2. *Second constraint requires*

$$\exp\{-z\} \leqslant x_2 \rightsquigarrow -z \leqslant \log x_2 \rightsquigarrow z \geqslant -\log x_2$$

3. *And, finally the third one restricts z to satisfy*

$$z \leqslant x_3$$

So, at $x = [x_1, x_2, x_3]$ we have

$$\mathbb{Z}(x) = \left(\left[-\infty, -\sqrt{-x_1}\right] \cup \left[\sqrt{-x_1}, +\infty\right]\right) \cap [-\log x_2, \infty] \cap [-\infty, x_3]$$

Generator Distribution: *Example*

- At $x = [-1, \exp\{2\}, 2]$, we have

$$\mathbb{Z}(x) = [-2, -1] \cup [1, 2] \rightsquigarrow \text{CDF}(x) = \int_{-2}^{-1} P(z)\mathrm{d}z + \int_{1}^{2} P(z)\mathrm{d}z$$

- At $x = [-1, 1, 2]$, we have

$$\mathbb{Z}(x) = [1, 2] \rightsquigarrow \text{CDF}(x) = \int_{1}^{2} P(z)\mathrm{d}z$$

- At $x = [a, 1, 2]$ with $a > 0$, we have

$$\mathbb{Z}(x) = \varnothing \rightsquigarrow \text{CDF}(x) = 0$$

Expressing $P(x)$ for all $x \in \mathbb{R}^3$ is quite cumbersome!

MLE Challenge: *Non-Computability of Likelihood*

Computability of Generator Distribution

With a non-invertible generator $G_{\mathbf{w}}$, it is challenging to characterize

$$\mathbb{Z}_{\mathbf{w}}(x) = \{z \in \mathbb{R}^m : G_{\mathbf{w}}(z) \leqslant x\}$$

and thus the output distribution $P_{\mathbf{w}}(x)$ is hard to be computed for all $x \in \mathbb{R}^d$

To train the generator by MLE $\rightsquigarrow$ *we need likelihood at all samples x^j*

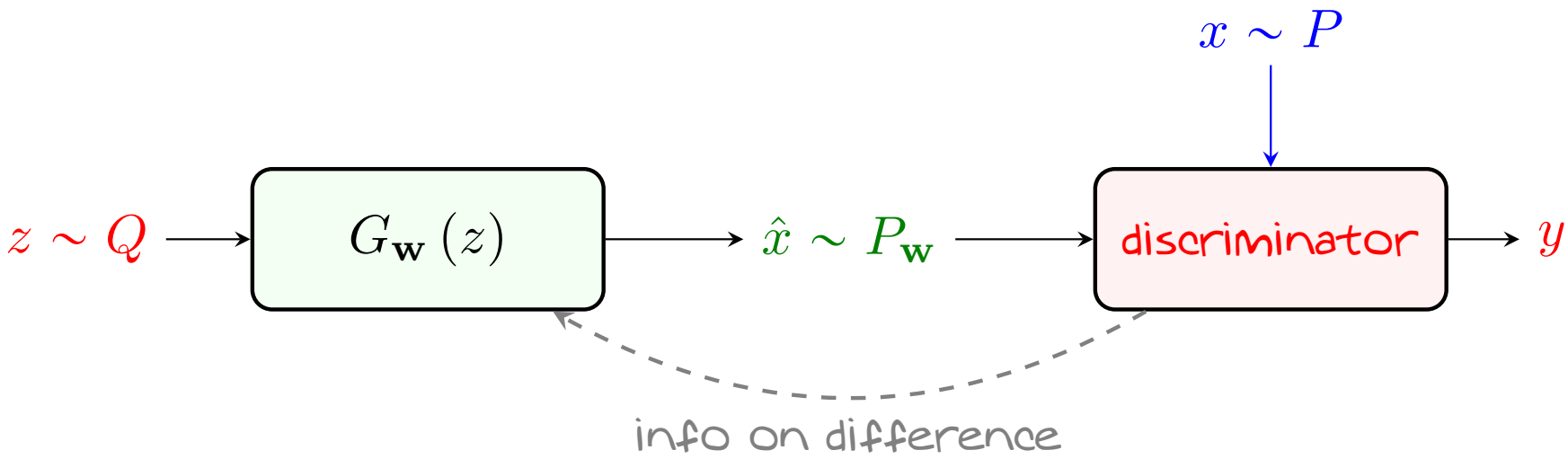
$$\hat{R}(\mathbf{w}) = -\frac{1}{n}\sum_j \boxed{\log P_{\mathbf{w}}(x^j)}$$

?!

Computability of Likelihood

With non-invertible generator, likelihood is not directly computable

Alternative Learning Approach: *Adversarial Architecture*



Discriminator

Discriminator is a binary classifier that classifies sample x as real or fake

To train with discriminator: *we can easily make a labeled dataset*

- *Data samples are labeled as real*
- *Generated samples are labeled as fake*

Dataset for *Adversarial Architecture*

From discriminator viewpoint: *we train a classifier on $2n$ labeled samples*

$$\mathbb{D} = \left\{ \boxed{\left(x^j, 1\right)}, \boxed{\left(\hat{x}^j, 0\right)} : j = 1, \ldots, n \right\}$$

real data sample

fake generated sample

We can compactly write the dataset as

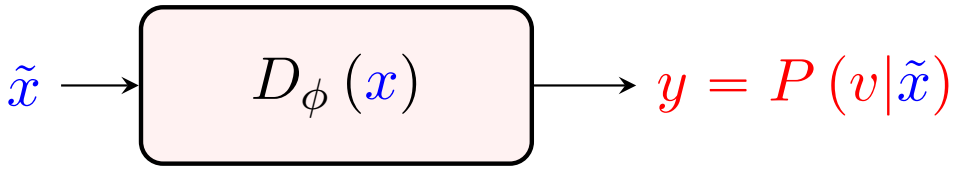
$$\mathbb{D} = \left\{ \left(\tilde{x}^j, v^j\right) : j = 1, \ldots, 2n \right\}$$

where the true label v is defined as

$$v = \begin{cases} 0 & x \text{ if fake} \equiv \tilde{x} = G_{\mathbf{w}}\left(z\right) \\ 1 & x \text{ if real} \equiv \tilde{x} = x \sim P\left(x\right) \end{cases}$$

Training *Computational Discriminator*

We now consider a computational discriminator: *a model* $D_\phi : \mathbb{R}^d \mapsto [0, 1]$, *e.g., NN with Sigmoid output, which classifies sample* x



Recall

This is a discriminative model! This explains the appellation ☺

This is a classical classification problem $\rightsquigarrow$ *we use cross-entropy loss*

$$\mathrm{CE}(y, v) = -v \log y - (1 - v) \log(1 - y)$$

$$= \begin{cases} -\log y & \text{if } \tilde{x} \text{ is real} \equiv v = 1 \\ -\log(1 - y) & \text{if } \tilde{x} \text{ is fake} \equiv v = 0 \end{cases}$$

Empirical Risk of *Adversarial Architecture*

To train on this labeled dataset $\rightsquigarrow$ *we minimize the empirical risk, i.e.,*

$$
\begin{aligned}
\hat{R}(\mathbf{w}, \phi) &= \frac{1}{2n} \sum_j \mathrm{CE}\left(y^j, v^j\right) \\
&= \frac{1}{2n} \sum_j \mathrm{CE}\left(D_\phi\left(\tilde{x}^j\right), v^j\right) \\
&= \frac{1}{n} \sum_j \mathrm{CE}\left(D_\phi\left(x^j\right), 1\right) + \frac{1}{n} \sum_j \mathrm{CE}\left(D_\phi\left(G_{\mathbf{w}}\left(z^j\right)\right), 0\right) \\
&= \hat{\mathbb{E}}_{x \sim P}\left\{\mathrm{CE}\left(D_\phi(x), 1\right)\right\} + \hat{\mathbb{E}}_{z \sim Q}\left\{\mathrm{CE}\left(D_\phi\left(G_{\mathbf{w}}(z)\right), 0\right)\right\} \\
&= \hat{\mathbb{E}}_{x \sim P}\left\{-\log D_\phi(x)\right\} + \hat{\mathbb{E}}_{z \sim Q}\left\{-\log\left(1 - D_\phi\left(G_{\mathbf{w}}(z)\right)\right)\right\} \\
&= -\left[\hat{\mathbb{E}}_{x \sim P}\left\{\log D_\phi(x)\right\} + \hat{\mathbb{E}}_{z \sim Q}\left\{\log\left(1 - D_\phi\left(G_{\mathbf{w}}(z)\right)\right)\right\}\right]
\end{aligned}
$$

Empirical risk depends on both generator and discriminator

Training by Min-Max Game: *Discriminator's Role*

To train the adversarial architecture *by the* empirical risk

$$\hat{R}(\mathbf{w}, \phi) = -\left[\hat{\mathbb{E}}_{x \sim P}\left\{\log D_{\phi}(x)\right\} + \hat{\mathbb{E}}_{z \sim Q}\left\{\log\left(1 - D_{\phi}\left(G_{\mathbf{w}}(z)\right)\right)\right\}\right]$$

we let the discriminator and generator play a game

Discriminator tries its best to correctly classify fake sample from true ones, i.e.,

$$\begin{aligned}\min_{\phi} \hat{R}(\mathbf{w}, \phi) &\equiv \max_{\phi} \boxed{\hat{\mathbb{E}}_{x \sim P}\left\{\log D_{\phi}(x)\right\} + \hat{\mathbb{E}}_{z \sim Q}\left\{\log\left(1 - D_{\phi}\left(G_{\mathbf{w}}(z)\right)\right)\right\}} \\ &\equiv \max_{\phi} \boxed{\mathcal{L}(\mathbf{w}, \phi)}\end{aligned}$$

After maximization, discriminator's best try gets the objective

$$\mathcal{L}^{\star}(\mathbf{w}) = \max_{\phi} \mathcal{L}(\mathbf{w}, \phi)$$

Training by Min-Max Game: *Generator's Role*

Generator tries its best to *confuse discriminator*

- *This way discriminator cannot distinguish between fake and true samples*
  - ↳ *This means that fake samples look the same as the true ones*
- *The generator can only play with its parameters* $\mathbf{w}$

Generator tries to make discriminator's best try as bad as possible, i.e.,

$$\min_{\mathbf{w}} \mathcal{L}^{\star}(\mathbf{w}) = \min_{\mathbf{w}} \max_{\phi} \mathcal{L}(\mathbf{w}, \phi)$$

Statistical Perspective

Generator minimizes discriminator's maximal likelihood $\rightsquigarrow$ *discriminator remains confused between generated and true samples even if it does its best*

GAN: *Generative Adversarial Network*

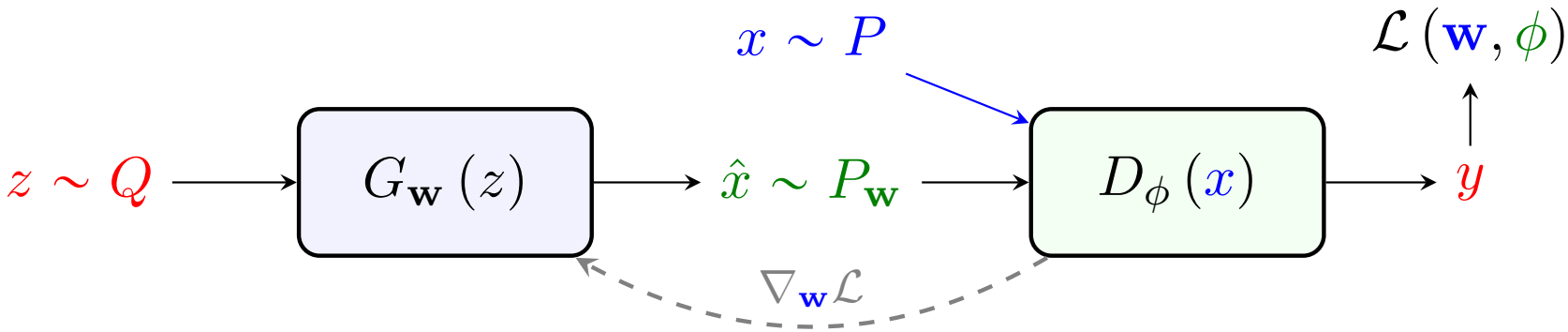
Generative Adversarial Network

GAN consists of a generator $G_{\mathbf{w}}$ and discriminator D_{ϕ} that are jointly learned through the min-max game

$$\min_{\mathbf{w}} \max_{\phi} \mathcal{L}(\mathbf{w}, \phi)$$

with objective function

$$\mathcal{L}(\mathbf{w}, \phi) = \hat{\mathbb{E}}_{x \sim P} \{\log D_{\phi}(x)\} + \hat{\mathbb{E}}_{z \sim Q} \{\log (1 - D_{\phi}(G_{\mathbf{w}}(z)))\}$$



GAN: *Training Loop*

`Train_vanillaGAN(`$\mathbb{D}$`:dataset):`

Initiate the generator $G_{\mathbf{w}}$ *and discriminator* D_{ϕ} *with some* $\mathbf{w}$ and ϕ
for *multiple epochs* **do**
  Sample a batch of data samples $\left\{x^j : j = 1, \ldots, n\right\}$ *from* $\mathbb{D}$
  Sample latents $\left\{z^j : j = 1, \ldots, n\right\}$ *using* Q *and compute* $\hat{x}^j \leftarrow G_{\mathbf{w}}\left(z^j\right)$
  for $\xi = 1, \ldots, \Xi$ **do**
    for $j = 1, \ldots, n$ **do**
      Compute $\mathcal{L}^j = \log D_{\phi}\left(x^j\right) + \log\left(1 - D_{\phi}\left(\hat{x}^j\right)\right)$
      Backpropagate over discriminator to compute $\nabla_{\phi}\mathcal{L}^j$
      if $\xi = \Xi$ **then** *Backpropagate over generator to compute* $\nabla_{\mathbf{w}}\mathcal{L}^j$
    end for
    Update ϕ *using* `Opt_avg` $\left\{-\nabla_{\phi}\mathcal{L}^j\right\}$
  end for
  Update $\mathbf{w}$ *using* `Opt_avg` $\left\{\nabla_{\mathbf{w}}\mathcal{L}^j\right\}$
end for
return *trained generator* $G_{\mathbf{w}}$

GAN Training: *Few Notes*

There are a few tricks to make training work

1. *We update discriminator for Ξ steps before updating generator once*
   - ↳ *This allows the inner maximization to slightly converge*

$$\min_{\mathbf{w}} \max_{\phi} \mathcal{L}(\mathbf{w}, \phi)$$

   - ↳ *Typical choices are $5 \leqslant \Xi \leqslant 10$*
2. *To update generator, we only need to compute derivative of second term*

$$\nabla_{\mathbf{w}} \mathcal{L}^j = \underbrace{\nabla_{\mathbf{w}} \log D_{\phi}\left(x^j\right)}_{0} + \nabla_{\mathbf{w}} \log\left(1 - D_{\phi}\left(G_{\mathbf{w}}\left(z^j\right)\right)\right)$$

   - ↳ *This gives very small gradients first and exploding later*
     - ↳ *First $D_{\phi}\left(G_{\mathbf{w}}\left(z^j\right)\right) \approx 0$ and later when it's fooled $D_{\phi}\left(G_{\mathbf{w}}\left(z^j\right)\right) \approx 1$*
     - ↳ *This is opposite of what we want!*
   - ↳ *Heuristic remedy is to use $-\nabla_{\mathbf{w}} \log D_{\phi}\left(G_{\mathbf{w}}\left(z^j\right)\right)$ instead*
     - ↳ *Some people call $\mathcal{L}_G^j = -\log D_{\phi}\left(G_{\mathbf{w}}\left(z^j\right)\right)$ generator loss*

GAN: *Sampling*

Sampling is exactly as in flow-based models

`Sample_GAN():`

1: *Sample latent as* $z \sim Q(z)$ `#Gaussian noise`
2: *Pass through generator* $x \leftarrow G_{\mathbf{w}}(z)$
3: **return** generated sample x

Though working with tricks, the vanilla GAN training is generally unstable

Wasserstein GAN addresses this issue by a very smart trick

To understand the trick used by Wasserstein GAN, we need to first learn the statistical interpretation of vanilla GAN training